Arduino-Based Environmental Monitoring System Using SCD30 CO_2 Sensor, SHT40 Sensor, and SD Card Data Logging

1. Introduction

Environmental monitoring plays a vital role in agriculture, laboratories, industrial facilities, buildings, and indoor air management. Parameters such as carbon dioxide (CO_2) concentration, temperature, and relative humidity directly influence human health, equipment performance, and plant growth. Continuous monitoring of these environmental variables enables timely decision-making, improves safety, and supports the maintenance of optimal environmental conditions (Moura.,2026).

Advances in low-cost sensors have made it possible to develop compact, accurate, and affordable environmental monitoring systems. Arduino-based platforms have become popular for research, education, and prototype development because they are open-source, inexpensive, easy to program, and compatible with numerous digital sensors (Tsebesebe.,2025). These characteristics make Arduino suitable for real-time data acquisition and environmental monitoring applications (Chan., 2021).

This project presents the design and implementation of a low-cost environmental monitoring system capable of measuring carbon dioxide concentration, air temperature, and relative humidity while continuously recording the collected data for later analysis. The developed system aims to provide an affordable and scalable solution for environmental monitoring in educational, laboratory, greenhouse, and indoor air quality applications.

1.2. Objective

To design and develop an Arduino-based environmental monitoring system capable of measuring carbon dioxide (CO_2) concentration, air temperature, and relative humidity in real time while recording the collected data onto a microSD card for subsequent analysis.

2.0. Hardware

2.1 Hardware Design

This project involved the design, assembly, programming, and testing of a portable environmental monitoring system based on an Arduino Uno microcontroller. The system was designed to continuously measure atmospheric carbon dioxide concentration, air temperature, and relative humidity while simultaneously storing the acquired data on a microSD memory card. System evaluation focused on verifying successful sensor communication, real-time data acquisition, and reliable data logging.

The environmental monitoring system consisted of an Arduino Uno microcontroller, an SCD30 carbon dioxide sensor, an SHT40 temperature and relative humidity sensor, a DFRobot microSD card module, a microSD memory card, jumper wires, and a USB cable for programming and power supply. Table 1 presents the bill of materials used in the project.

2.2 Bill of Materials

Table 1. Bill of materials for the Arduino-based environmental monitoring system.

Item	Component	Arduino Connection Pins	Communication Interface	Operating Voltage	Function	Qty.	Approx. Cost (USD)	Product Reference
1	Geekcreit Arduino Uno R3-Compatible Microcontroller (ATmega328P)	USB Type-B	USB, I^2C, SPI, UART	5 V	Central processing unit responsible for sensor interfacing, data acquisition, processing and data logging.	1	10.0	https://www.banggood.com/GEEKCREIT-UNO-R3-ATmega328P-Development-Board-Compatible-with-Arduino-p-964163.html
2	Sensirion SCD30 CO_2, Temperature and Relative Humidity Sensor	VCC→5 V; GND→GND; SDA→SDA; SCL→SCL	I^2C	5.5 V	Measures atmospheric CO_2 concentration, air temperature and relative humidity.	1	41.13	https://sensirion.com/products/catalog/SCD30
3	Adafruit SHT40 Temperature and Relative Humidity Sensor	VCC→5 V; GND→GND; SDA→A4 (SDA); SCL→A5 (SCL)	I^2C	5 V*	Measures air temperature and relative humidity.	1	5.95	https://www.adafruit.com/search?q=sht40
4	DFRobot Fermion MicroSD Card Module (DFR0229)	VCC→5 V; GND→GND; CS→D10; MOSI→D11; MISO→D12; SCK→D13	SPI	3.3-5 V	Stores environmental measurements on a microSD card in CSV format.	1	5.20	https://www.dfrobot.com/product-875.html
5	SanDisk Ultra microSD Card (16 GB, Class 10)	Inserted into microSD module	SPI (via module)	—	Stores recorded environmental measurements.	1	8.00	https://www.sandisk.com/products/memory-cards/microsd-cards/sandisk-ultra-uhs-i-microsd
6	Jumper Wires	Power and signal connections	Electrical	—	Provides electrical connections between components.	1 set	4.00	https://www.sparkfun.com/products/11026
7	USB Type-A to Type-B Cable	USB Type-B	USB 2.0	5 V	Supplies power and enables programming and serial communication.	1	3.00	https://www.startech.com/en-us/cables/usb2hab1m
8	Computer (Windows 11)	USB	USB	—	Programming, serial monitoring and data retrieval.	1	—	Existing laboratory equipment
	Total Approximate cost						77.28	

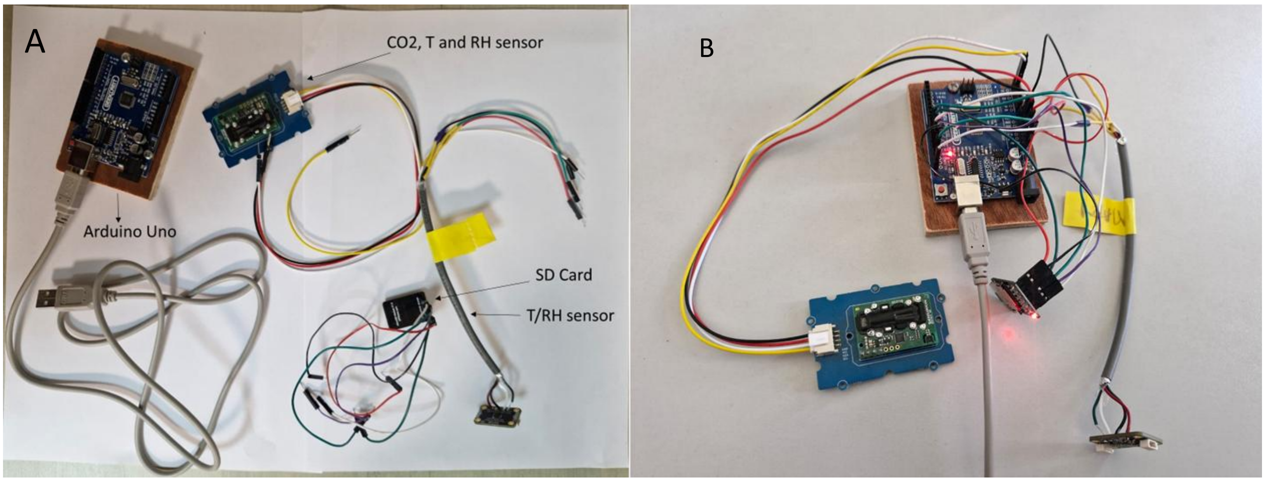

Figure 1. shows (A) Individual hardware components and (B) assembled prototype of the Arduino-based environmental monitoring system.

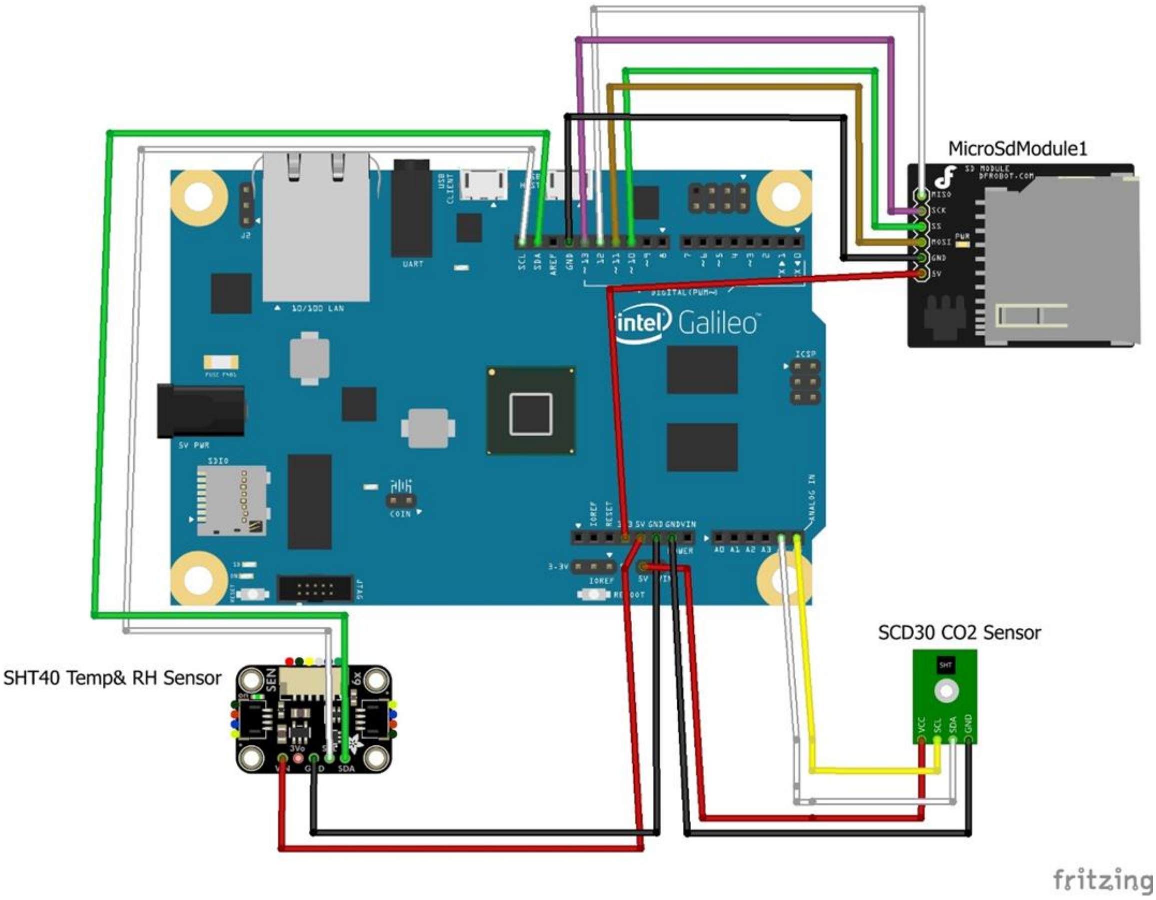

Figure 2. Physical wiring diagram of the Arduino based monitoring system

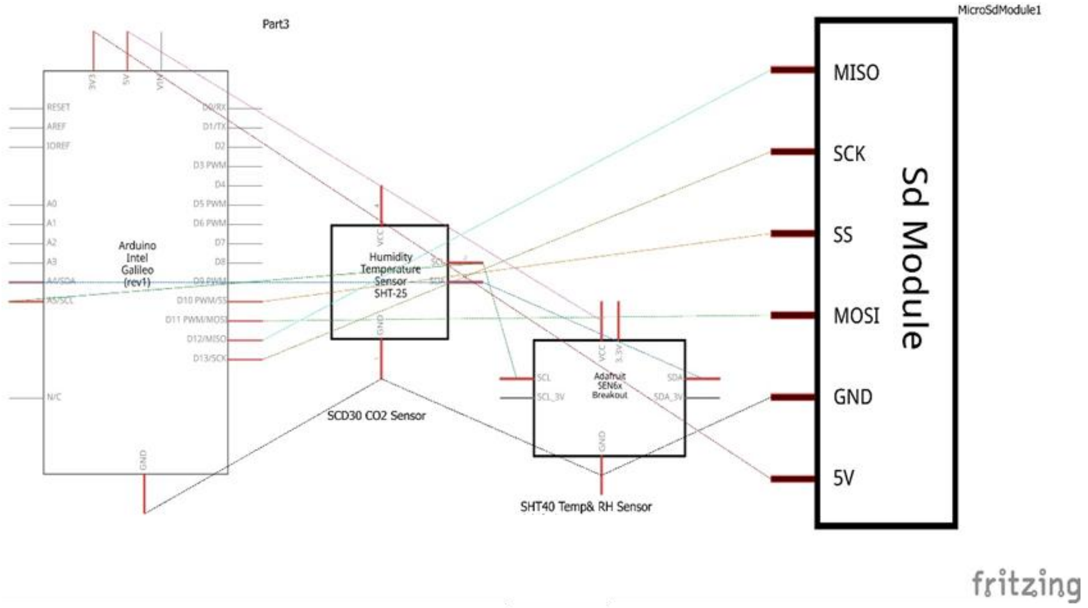

Figure 3. System wiring schematic showing the electrical connections between the Arduino, SCD30 sensor, SHT40 sensor, and microSD card module.

3.0 Software Development

3.1 Software flowchart

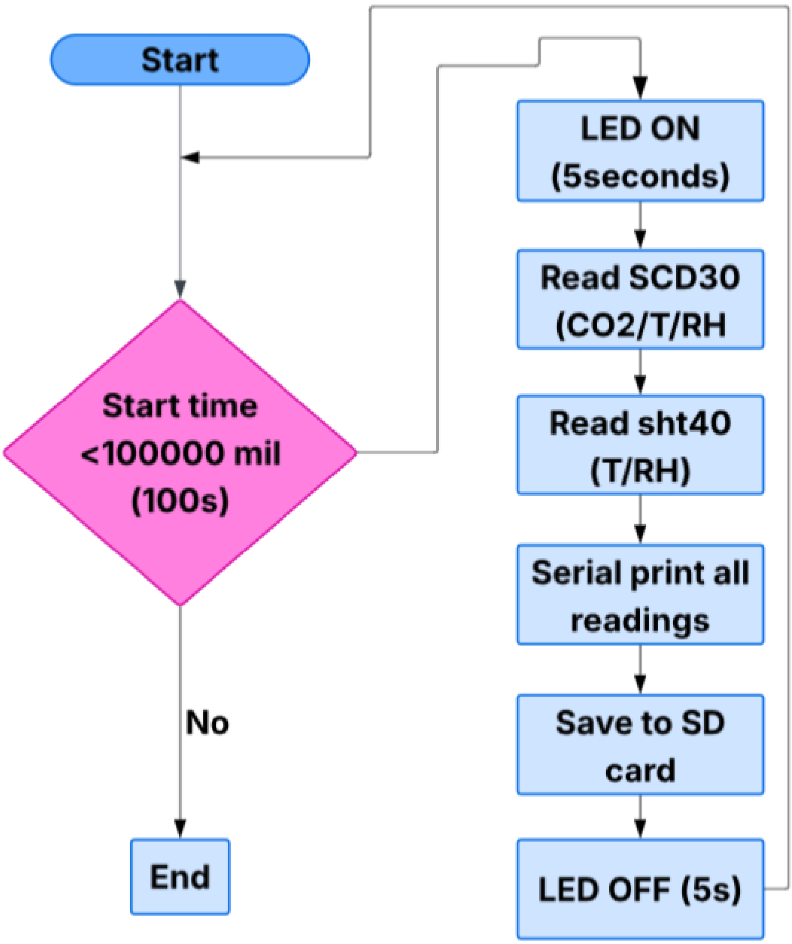

Figure 4. Software flowchart illustrating the sequence of operations performed by the Arduino-based environmental monitoring system

3.2 Program Code

The Arduino program was developed using the Arduino IDE to control sensor communication, acquire environmental measurements, display real-time data on the Serial Monitor, and store measurements on the microSD card in CSV format. The program initializes the required libraries, establishes communication with the SCD30 and SHT40 sensors through the I²C protocol, configures the microSD card through the SPI interface, and continuously executes the measurement and data logging cycle during the monitoring period.

```
#include <Wire.h>
#include <SPI.h>
#include <SD.h>
#include <Adafruit_SCD30.h>
#include <Adafruit_SHT4x.h>

Adafruit_SCD30 scd30;
Adafruit_SHT4x sht4;

const int chipSelect = 10;
unsigned long startTime;

void setup() {

  Serial.begin(9600);

  scd30.begin();
  sht4.begin();

  SD.begin(chipSelect);

  pinMode(LED_BUILTIN, OUTPUT);

  File dataFile = SD.open("data.csv", FILE_WRITE);

  if (dataFile) {
    dataFile.println("Time,CO2,SCD30_Temp,SCD30_RH,SHT_Temp,SHT_RH");
    dataFile.close();
  }

  startTime = millis();
}

void loop() {

  if (millis() - startTime < 100000) {

    // LED ON
    digitalWrite(LED_BUILTIN, HIGH);
    delay(5000);

    // Read SCD30
    scd30.read();

    // Read SHT40
    sensors_event_t humidity, temp;
```

```
    sht4.getEvent(&humidity, &temp);

    // Print to Serial Monitor
    Serial.print("CO2: ");
    Serial.print(scd30.CO2);

    Serial.print(" | SCD30 T: ");
    Serial.print(scd30.temperature);

    Serial.print(" | SCD30 RH: ");
    Serial.print(scd30.relative_humidity);

    Serial.print(" | SHT T: ");
    Serial.print(temp.temperature);

    Serial.print(" | SHT RH: ");
    Serial.println(humidity.relative_humidity);

    // Save to SD card
    File dataFile = SD.open("data.csv", FILE_WRITE);

    if (dataFile) {

      dataFile.print(millis());
      dataFile.print(",");

      dataFile.print(scd30.CO2);
      dataFile.print(",");

      dataFile.print(scd30.temperature);
      dataFile.print(",");

      dataFile.print(scd30.relative_humidity);
      dataFile.print(",");

      dataFile.print(temp.temperature);
      dataFile.print(",");

      dataFile.println(humidity.relative_humidity);

      dataFile.close();
    }

    // LED OFF
    digitalWrite(LED_BUILTIN, LOW);
    delay(5000);

  }
  else {

    Serial.println("Program stopped.");

    while (1);
  }
}
```

4.0 Results and Discussion

4.1 Real-Time Serial Monitor Output

The Arduino-based monitoring system successfully acquired real-time environmental measurements from the Sensirion SCD30 and SHT40 sensors and displayed the recorded data on the Arduino Serial Monitor. The output included atmospheric CO_2 concentration measured by the SCD30 sensor, together with air temperature and relative humidity measurements obtained from both the SCD30 and SHT40 sensors. As shown in Figure 5. The displayed readings confirmed successful communication between the Arduino microcontroller and the sensors, demonstrating reliable real-time data acquisition and visualization throughout the 100-second monitoring period.

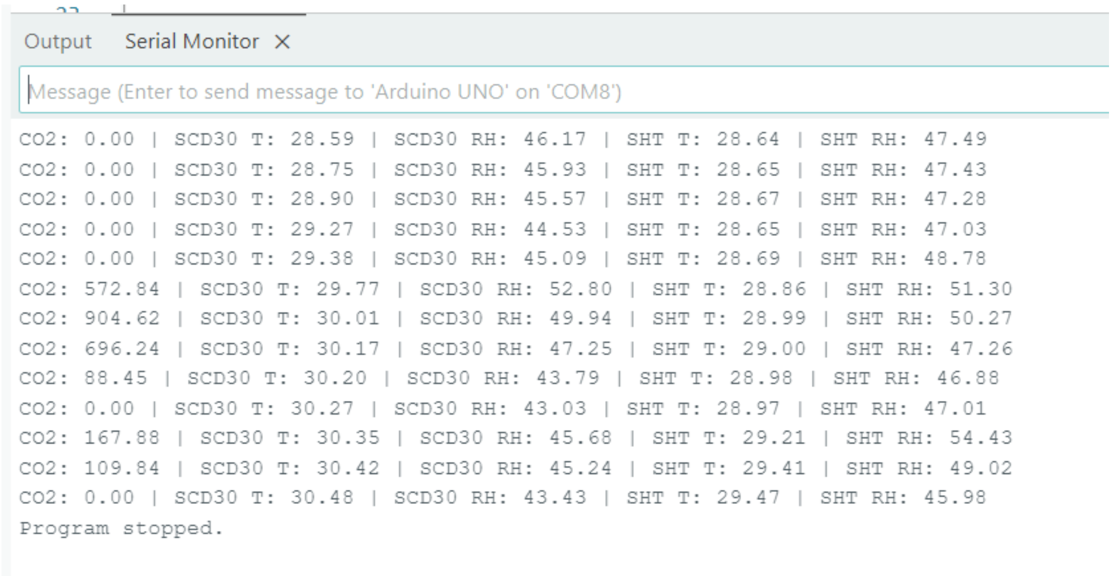

Figure 5. Real-time sensor measurements displayed on the Arduino Serial Monitor, showing atmospheric CO_2 concentration, air temperature, and relative humidity recorded from the SCD30 and SHT40 sensors during system operation.

4.2 MicroSD Card Data Logging

The environmental measurements acquired during system operation were successfully stored on the microSD card in CSV format. Each record contained the elapsed time, atmospheric CO_2 concentration, SCD30 temperature, SCD30 relative humidity, SHT40 temperature, and SHT40 relative humidity. The successful generation of the CSV file demonstrated reliable data logging and confirmed the system's ability to preserve environmental measurements for subsequent analysis.

Time	CO2	SCD30_Temp	SCD30_RH	SHT_Temp	SHT_RH
5102	0	29.27	44.53	28.65	47.03
15151	0	29.38	45.09	28.69	48.78
25201	572.84	29.77	52.8	28.86	51.3
35252	904.62	30.01	49.94	28.99	50.27
45309	696.24	30.17	47.25	29	47.26
55360	88.45	30.2	43.79	28.98	46.88
65409	0	30.27	43.03	28.97	47.01
75459	167.88	30.35	45.68	29.21	54.43
85511	109.84	30.42	45.24	29.41	49.02
95559	0	30.48	43.43	29.47	45.98

Figure 6. Sample environmental data recorded on the microSD card in CSV format, showing the elapsed time, atmospheric CO_2 concentration, SCD30 temperature, SCD30 relative humidity, SHT40 temperature, and SHT40 relative humidity measured during system operation.

5.0 Conclusion.

The project demonstrated that a low-cost Arduino-based platform can be used to develop a functional environmental monitoring system capable of acquiring, displaying, and storing atmospheric CO_2 concentration, air temperature, and relative humidity data. The prototype provides a simple, and scalable solution that can be adapted for educational purposes, laboratory monitoring, greenhouse applications, and indoor air quality monitoring. Future work may include the integration of additional environmental sensors, wireless communication for remote monitoring, cloud-based data storage, and long-term field validation to further enhance system functionality, reliability, and performance.

6.0. References

Moura, P. C., Alieksieiev, V., Domingues, H., Pessanha, S., & Vassilenko, V. (2026). An Innovative Multi-Parameter Environmental Sensor System for Real-Time Indoor Air Quality Monitoring in Industrial Facilities. *Sustainability*, *18*(14), 7080. https://doi.org/10.3390/su18147080

Tsebesebe, N. T., Mpofu, K., Sivarasu, S., & Mthunzi-Kufa, P. (2025). Arduino-based devices in healthcare and environmental monitoring. *Discover Internet of Things*, *5*(1), 46.

Chan, K., Schillereff, D. N., Baas, A. C., Chadwick, M. A., Main, B., Mulligan, M., ... & Thompson, J. (2021). Low-cost electronic sensors for environmental research: Pitfalls and opportunities. *Progress in Physical Geography: Earth and Environment*, *45*(3), 305-338.